

MSc DISSERTATION, 2026

# Breast Cancer Ultrasound Classification Using Machine Learning

Auditable evaluation and research dashboard

LOCAL COHORT EVIDENCE

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# Model scores and evidence strength are separate

## The modelling problem

- Ultrasound varies with operator, device, acquisition settings, anatomy, and speckle.
- Rectangular images can be geometrically distorted by direct square resizing.
- Small public datasets make leakage and undocumented preprocessing unusually consequential.

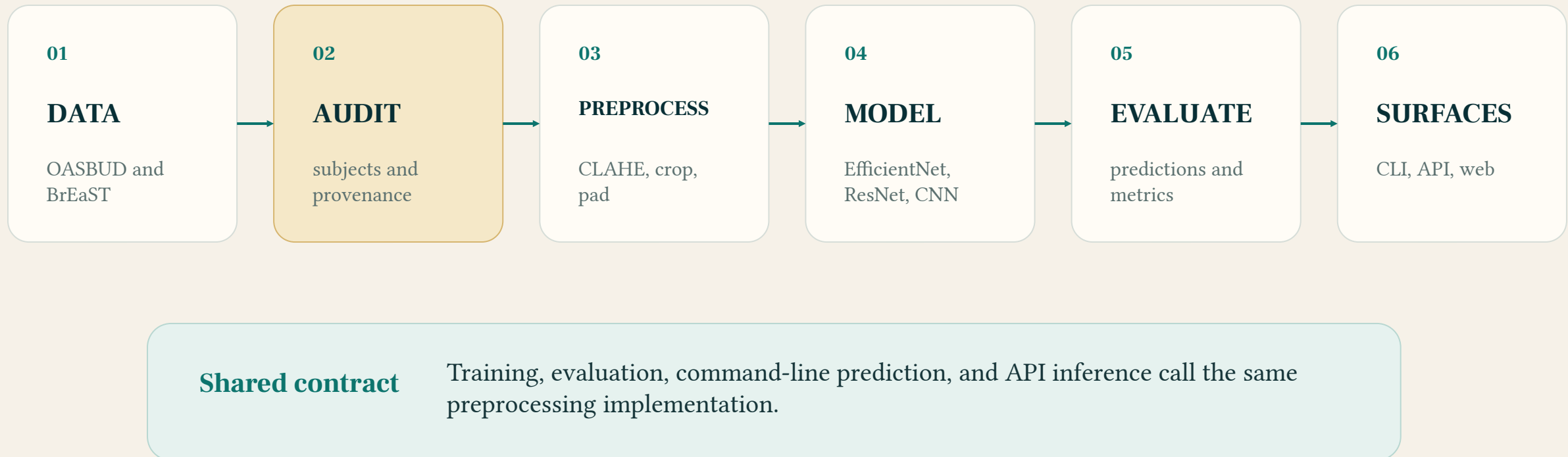
## Engineering evidence

The code path executes consistently. Its artifacts are reproducible and auditable.

## Generalisation evidence

The repaired local cohort has subject separation. External and clinical validation remain outside this study.

# One controlled path connects the data to every output



# Validated datasets pass the subject-level split audit

| Dataset         | Train | Validation | Test | Audit finding                                |
|-----------------|-------|------------|------|----------------------------------------------|
| BUSI derivative | n/a   | n/a        | n/a  | Excluded: provenance cannot be reconstructed |
| OASBUD          | 138   | 30         | 30   | Pass: no subject crosses a partition         |
| BrEaST          | 164   | 33         | 32   | Pass: no subject crosses a partition         |
| Validated       | 302   | 63         | 62   | Pass: local cohort evidence                  |

Release gate

The BUSI derivative cannot prove patient separation or reproduce published curation checks, so it remains outside the experiment.

# Preprocessing is configurable, shared, and testable

01

## DIRECT RESIZE

Rectangular image is resized directly to 224×224.

Comparator. Scaling differs across the two axes.

02

## CENTRE CROP

A centred square is selected before resizing.

Comparator. Peripheral context may be removed.

03

## MASK / FALLBACK ✦ PAD

Mask box or central 80%, then margin and reflected square padding.

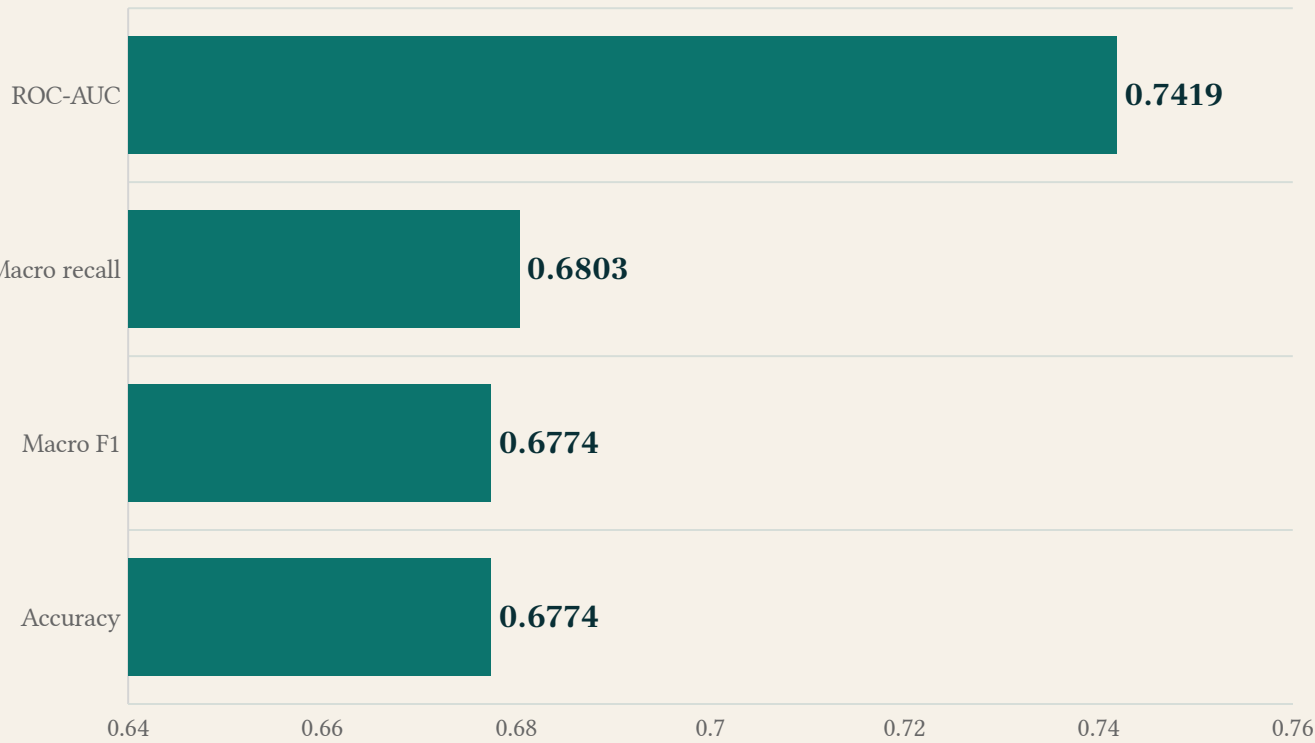
Default research path. The crop retains its aspect ratio.

### Controlled ablation rule

Hold the subject partition, architecture, training budget, and random seeds fixed. Change one preprocessing factor at a time.

Training augmentation uses horizontal flips and rotations up to 15°. Vertical flips are excluded because image depth has acquisition meaning.

# Audited test performance



CONFUSION MATRIX

| Actual / predicted | Benign | Malignant |
|--------------------|--------|-----------|
| Benign             | 21     | 12        |
| Malignant          | 8      | 21        |

62 test images, EfficientNet-B0

Local cohort evidence only. Clinical validation is outside scope.

# The project now leaves a verifiable evidence trail

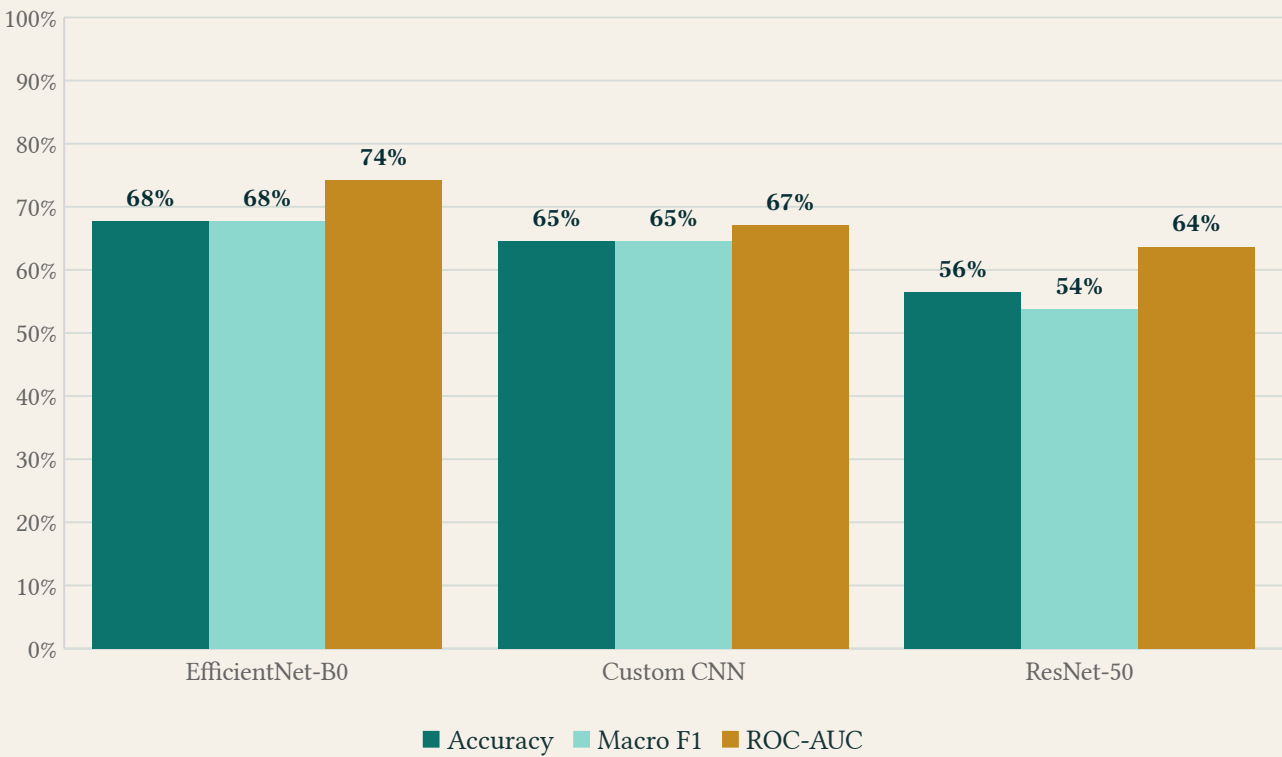
- Subject-level audit is regenerated before training. Failed audits stop by default.
- Python, NumPy, PyTorch, and CUDA seeds are set and checkpoint configuration is saved.
- Evaluation writes one row per image plus metrics, figures, audit status, and checksums.
- Mask discovery, crop dimensions, audit failure, and the non-clinical API boundary have regression tests.
- The dissertation, notebook, slides, README, and web interface use the same current evidence status.

## RELEASE ARTIFACTS

**data\_audit.json**  
**predictions.csv**  
**metrics.json**  
**extended\_evaluation.json**  
**report figures**

Generated, inspectable, and tied to one evaluation run.

# Architecture comparison on the audited test split



| Model           | Accuracy 95% CI  | ROC-AUC 95% CI   |
|-----------------|------------------|------------------|
| EfficientNet-B0 | 0.5645 to 0.7903 | 0.6217 to 0.8616 |
| Custom CNN      | 0.5323 to 0.7581 | 0.5330 to 0.7979 |
| ResNet-50       | 0.4355 to 0.6935 | 0.5054 to 0.7701 |

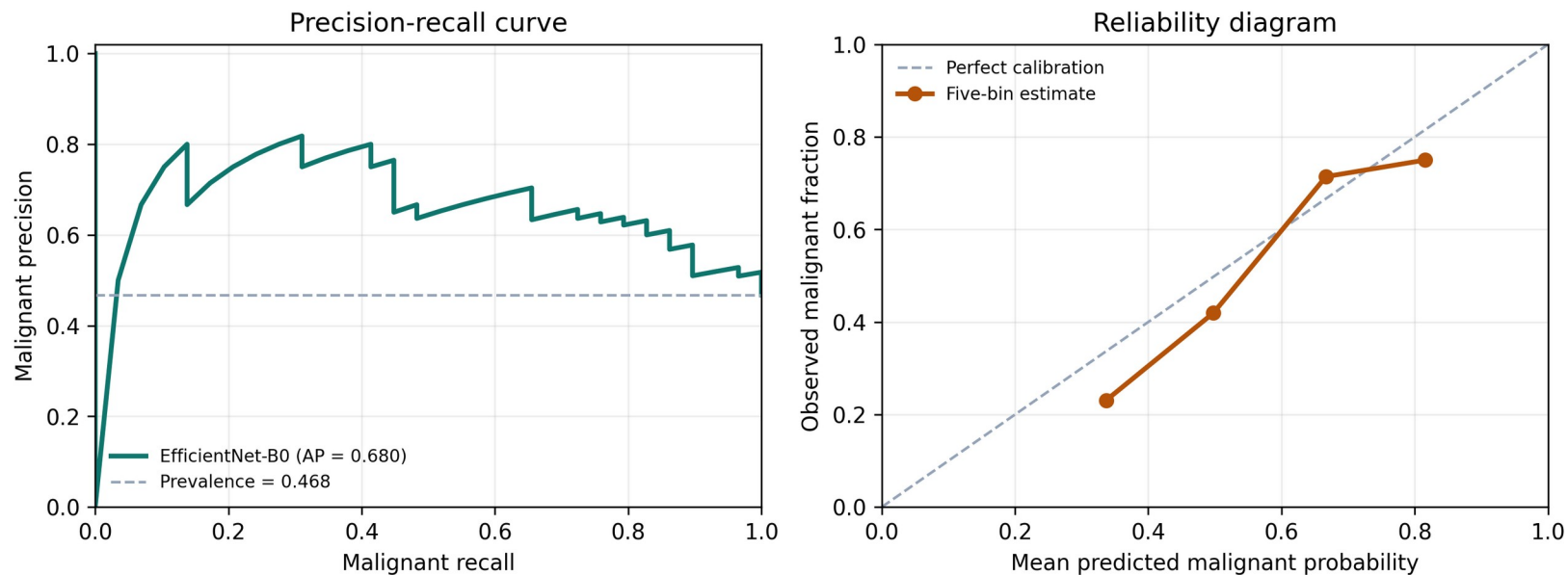
## Intervals overlap

The ranking is descriptive. The data do not support a superiority or breakthrough claim.

Source: outputs/model\_benchmark.json, one training seed and 62 test images

# Discrimination and calibration metrics

Threshold and calibration diagnostics from 62 stored test predictions



AVERAGE PRECISION

**0.6803**

BRIER SCORE

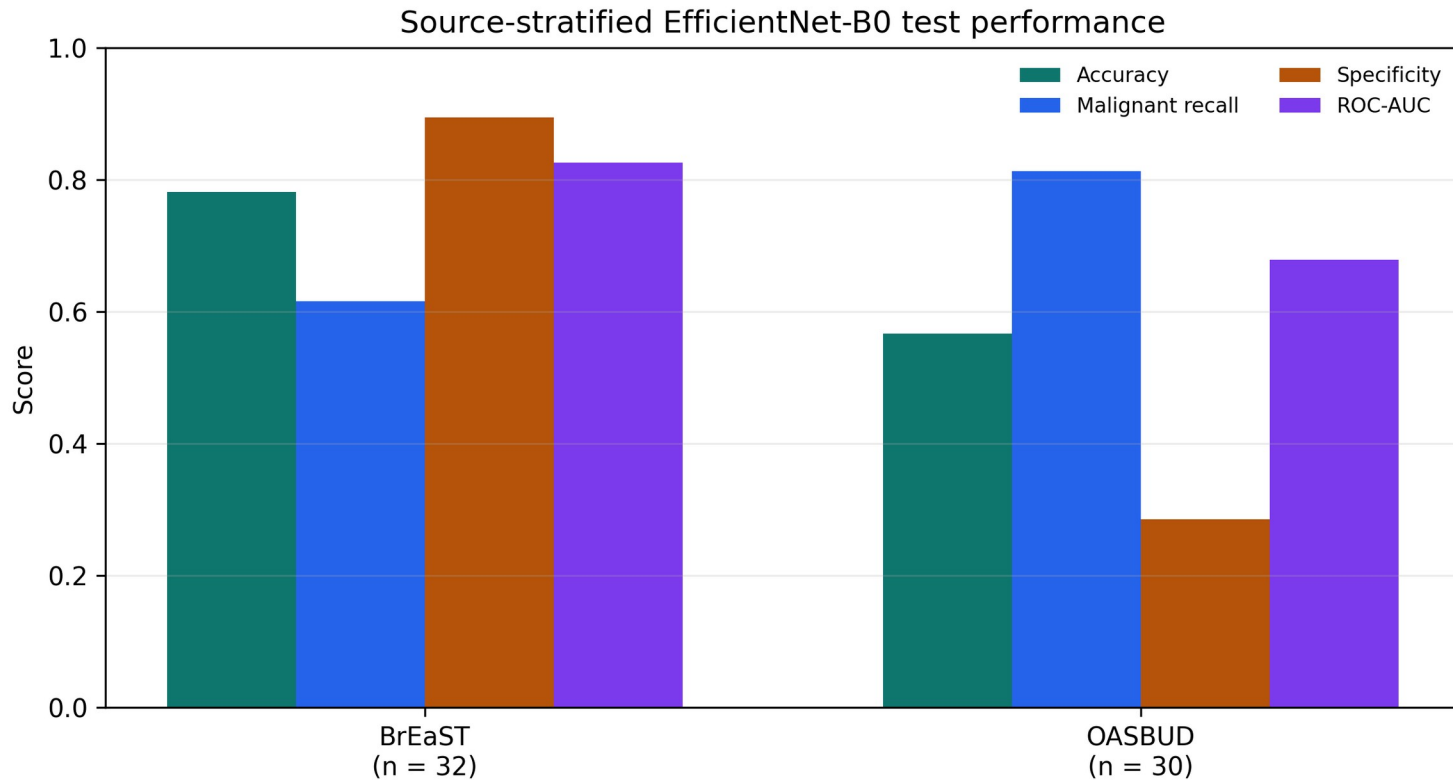
**0.2144**

EXPECTED CALIBRATION ERROR

**0.0652**

Five bins, descriptive test-set diagnostics

# Errors differ between BrEaST and OASBUD



| Source | n  | Accuracy | FP | FN |
|--------|----|----------|----|----|
| BrEaST | 32 | 0.7813   | 2  | 5  |
| OASBUD | 30 | 0.5667   | 10 | 3  |

## Interpretation boundary

Source, case mix, acquisition, reconstruction, and sample size are confounded. The plot cannot rank dataset quality.

# Six pages document the complete workflow

OncoVision  
Research workspace

Single image

Noise experiment

Batch evaluation

Evidence record

Dataset explorer

Documentation

Running inference...

Loaded model

Evidence

MLC dissemination prototype

custom\_cnn

Provisional

Maxwell David Darmstadter - 2026

Single-image inspection

Trace one image through preprocessing, classification, and visual explanation.

Model

Geometry

CLAHE

Provisional evidence All included datasets passed identifier and cross-partition checks. Review evidence

Inspect a model output

Accepted files: PNG, JPEG, or TIFF up to 20 MB. Results are research artifacts, never a diagnosis.

Choose image

Load labelled example

Compare available models

Source image

case046.png - known label malignant

Grad-CAM overlay

Grad-CAM

Processed

CLAHE

Raw

OncoVision  
Research workspace

Single image

Noise experiment

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Model ready

Loaded model

Evidence

MLC dissemination prototype

custom\_cnn

Provisional

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Noise experiment

Compare a seeded synthetic perturbation across documented preprocessing paths.

Model

Geometry

CLAHE

Provisional evidence All included datasets passed identifier and cross-partition checks. Review evidence

Seeded perturbation experiment

Compare one image before and after a reproducible synthetic perturbation. This is a demonstration, not a cohort robustness result.

Load labelled example

Run experiment

Noise type

Intensity

QOS

Comparator

Seed

Speckle

Direct resize

42

OncoVision  
Research workspace

Single image

Noise experiment

Batch evaluation

Evidence record

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Documentation

Model ready

Loaded model

Evidence

MLC dissemination prototype

custom\_cnn

Provisional

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Batch evaluation

Review multiple model outputs without assigning clinical priority.

Model

Geometry

CLAHE

Provisional evidence All included datasets passed identifier and cross-partition checks. Review evidence

Batch model evaluation

Process multiple images or load a labelled folder sample. No clinical prioritisation is performed.

Choose images

Load labelled cohort

Export CSV

Processed

12

Benign output

0

Malignant output

12

Mean softmax score

64.16%

OncoVision  
Research workspace

Single image

Noise experiment

Batch evaluation

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Model ready

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Evidence

MLC dissemination prototype

custom\_cnn

Provisional

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Evidence record

Read generated metrics together with the subject-level split audit.

Model

Geometry

CLAHE

Provisional evidence All included datasets passed identifier and cross-partition checks. Review evidence

Evidence record

One generated evaluation record, presented together with its split-integrity limitations.

Refresh files

Subject-level split audit

No cross-partition subject identifiers were found in the audited cohort.

BUSI

0 images in current folders

OASBUD

188 images in current folders

BREAST

229 images in current folders

OncoVision  
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Evidence

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custom\_cnn

Provisional

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Dataset explorer

Inspect local image folders and their current labels.

Model

Geometry

CLAHE

Provisional evidence All included datasets passed identifier and cross-partition checks. Review evidence

Local dataset explorer

Inspect folder labels and images. Folder membership does not certify subject independence.

Dataset

Partition

BREAST

Training

Showing 24 of 164 images. Counts reflect folder contents, not independent subjects.

Image 1

Image 2

Image 3

Image 4

OncoVision  
Research workspace

Single image

Noise experiment

Batch evaluation

Evidence record

Dataset explorer

Documentation

Model ready

Loaded model

Evidence

MLC dissemination prototype

custom\_cnn

Provisional

Maxwell David Darmstadter - 2026

Documentation

Read the submission-facing project record directly from source.

Model

Geometry

CLAHE

Provisional evidence All included datasets passed identifier and cross-partition checks. Review evidence

System guide and project record

Use boundaries, interface map, literature audit, and submission-facing documentation served from the repository.

APPROPRIATE USE

Research inspection

Reproduce local model outputs

Inspect preprocessing and Grad-CAM

UNSUPPORTED USE

Clinical decisions

No diagnosis or patient triage

No claim of external generalisation

EVIDENCE BOUNDARY

Audited local cohort

BREAST and OASBUD only

BUSI derivative excluded from validation

The dashboard documents research outputs. It does not provide a clinical service

## CONCLUSION

# A complete research system with an honest evidence boundary

### Defensible now

- Shared preprocessing and inference implementation
- Auditable evaluation artifacts and release checks
- Patient-separated local checkpoint with uncertainty estimates

### Claim boundary

- Clinical effectiveness or diagnostic safety
- External validation beyond the local combined cohort
- Statistically proven superiority over other architectures

Product repository: [github.com/DavidAkerele/breast-cancer-ultrasound-classification-ml](https://github.com/DavidAkerele/breast-cancer-ultrasound-classification-ml)

## Questions

No diagnostic, BI-RADS, or management inference is authorised by this prototype.